Efficient, Robust, and Accurate CNN Predictor for Neuronal Activation in Directional Deep Brain Stimulation

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***Abstract*— The programming of clinical deep brain stimulation (DBS) systems involves numerous combinations of stimulation parameters, such as stimulus amplitude, pulse width, and frequency. As more complex electrode designs, such as directional electrodes, are introduced, the traditional trial-and-error approach to manual DBS programming becomes increasingly impractical. Visualization of the volume of tissue activated (VTA) can assist in selecting stimulation parameters by showing the direct effects of DBS on neural tissue. However, the standard method for VTA calculation, which involves modeling biological nerve fibers, is highly time-consuming and limits clinical applicability. In this study, we used finite element models (FEM) of implanted DBS systems to compute electric fields and obtained a large dataset of axonal responses under electrical stimulation using multicompartment cable models. We then trained a convolutional neural network (CNN) to replace the cable models. The CNN model's performance in calculating VTA was evaluated across various electrode configurations and stimulation parameters, and compared with existing activation function (AF) methods. The CNN model achieved a mean absolute error (MAE) of 0.032V in predicting nerve fiber activation thresholds, demonstrating greater stability and accuracy in VTA prediction compared to the AF method. Additionally, the CNN reduced computation time by five orders of magnitude compared to standard axonal modeling methods. We demonstrate that the CNN-based neural fiber predictor can quickly, accurately, and robustly predict neural activation responses to DBS, thereby improving the efficiency of DBS programming.**

***Index Terms*— Convolutional neural network, directional deep brain stimulation, neuronal cable model, volume of tissue activated.**

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I. Introduction

DEEP brain stimulation (DBS) is an effective clinical method for treating various neurological disorders and has been approved by the U.S. Food and Drug Administration (FDA) for the treatment of essential tremor [1], Parkinson's disease [2], epilepsy [3], obsessive-compulsive disorder, and dystonia. Ongoing research is exploring the use of DBS for treating depression, Tourette syndrome, and many other neurological and psychiatric conditions [4]. As a precise neuromodulation therapy, the efficacy of DBS depends on the accurate implantation of electrodes at the target site and effective post-operative programming [5]. However, deviations between the implanted electrode position and the target site are inevitable [6]. To guide the stimulation current accurately to the target area while avoiding stimulation of unnecessary regions, directional electrodes have been developed [7]. These more complex electrodes offer advantages such as improving efficacy, reducing side effects, and conserving energy, but they also introduce greater challenges for clinical programming.

The programming of DBS systems involves the combination of multiple stimulation parameters, such as pulse width, amplitude, and frequency. A standard four-contact ring electrode offers over 25,000 possible combinations of stimulation parameters and 65 electrode contact configurations [8]. Selecting the optimal stimulation parameters from this vast array typically requires an iterative trial-and-error process based on the patient's response, relying heavily on the clinician's experience and being highly time-consuming. This process can also cause discomfort for the patient. To determine personalized parameters for each patient, multiple post-operative follow-up sessions, each lasting several hours, are often required [9]. This approach may be somewhat effective with standard ring electrodes, but with 8-contact directional electrodes, the number of possible configurations increases by 16 times, making manual programming extremely difficult.

To simplify and accelerate the DBS programming process, researchers have developed various computational models to predict DBS effects [9], [10], [11], [12]. Advanced models often integrate patient-specific diffusion-weighted imaging data with finite element models of implanted DBS systems to predict axonal pathway responses to stimulation [13], [14].

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In this study, we focus on the commonly used volume of tissue activated (VTA) model. By visualizing the patient's brain image and the VTA under specific stimulation parameters [15], clinicians can thoroughly explore the stimulation parameter space and assess the effects of stimulation in the computer, ultimately identifying the optimal parameter settings [16], [17], [18], thereby reducing the burden on both clinicians and patients. However, the standard VTA calculation method requires constructing a multi-compartment cable model of the axon [14] and solving the dynamic equations that govern axonal circuits, which is computationally intensive and time-consuming, making it unsuitable for real-time clinical calculations. Consequently, alternative methods that can compute VTAs more quickly have been developed, with most aiming to accurately replicate the VTAs generated by the axonal model method [19], [20].

The activation function (AF) [21] is a simplified method for calculating axonal responses to electrical stimulation. It uses the second-order difference of the potential at the nodes of Ranvier to predict activation thresholds and employs lookup tables to precompute prediction functions for each electrode configuration. Further work [22] has utilized weighted sums of these activation functions to achieve high-precision predictions while maintaining low computational costs. However, these models sometimes fail to accurately capture the complex nonlinear behavior of the cable model [23]. The AF-based method [20] performs poorly and exhibits instability under certain stimulation parameters.

In recent years, researchers have increasingly applied machine learning to calculate VTA or predict axonal responses. Chaturvedi [17] approximated the VTA as ellipse and developed an artificial neural network to predict the dimensional parameters of the ellipse. However, this approach is not suitable for predicting irregularly shaped VTAs generated by more complex electrodes, such as directional electrodes. Golabek [23] also developed an artificial neural network to achieve fast and accurate predictions of axonal activation and evaluated its performance under monopolar stimulation with ring electrodes. However, the study lacked discussion of the model's performance under more complex electric fields, such as those generated by bipolar stimulation or directional electrodes. In this study, we used a convolutional neural network (CNN) to directly predict the response of single axons to electrical stimulation, and evaluated its performance across a range of clinically relevant electrode configurations and stimulation parameters for both ring and directional electrodes. We proposed a high-performance CNN that can serve as a replacement for the multi-compartment cable model, providing fast, accurate, and robust predictions of axonal activation thresholds, and enabling precise VTA calculations.

II. Materials And Methods

A. Finite Element Model

We used COMSOL Multiphysics (version 6.1) to create finite element models of the implanted DBS electrodes (Lead-1 and Lead-2). Lead-1 has a diameter of 1.3 mm and consists of four ring-shaped stimulation contacts, with an inter-contact spacing of 0.5 mm. Lead-2 also has a diameter of 1.3 mm, featuring two conventional ring contacts and two sets of directional contacts, totaling eight contacts arranged in a 1-3-3-1 configuration, with an inter-contact spacing of 0.5 mm. The electrodes were positioned at the center of a homogeneous cubic brain tissue model (100 mm $\times$ 100 mm $\times$ 100 mm) with a conductivity of 0.1 S/m [20]. The conductivity of the electrode's insulation and contacts was set to 1×10^{-10} S/m and 1×106 S/m, respectively [24]. The electrode-tissue interface formed after electrode implantation was not considered in this model. We employed the AC/DC module and the current interface to solve the Laplace equation, predicting the static electric field generated by the electrodes within the homogeneous brain tissue. For monopolar stimulation, the active electrode contact was set to -1.0 V, while the tissue boundary was set to 0.0 V. For bipolar stimulation, the cathodal active contact was set to -0.5 V and the anodal contact to 0.5 V.

B. Axon Model

In the NEURON simulation environment (version 8.2.0) [25], we created a 5.7 μm diameter MRG multi-compartment double-cable axon model [26], which is the gold standard for simulating the response of mammalian myelinated axons to electrical stimulation [14], [27]. The axon model was arranged orthogonally to the electrode with a spacing of 0.4 mm. To reduce computational cost while preserving the potential distribution characteristics under different electrode configurations, for the ring electrode (Lead-1) with a symmetric electric field, we simulated 1,250 axons in a single plane. For the directional electrode (Lead-2), depending on whether the active contacts were on the same side, different sides with monopolar stimulation, or different sides with bipolar stimulation, we selected 2, 3, or 5 computation planes, respectively, with 620 axons per plane, with sparser sampling in regions farther from the electrode (In the calculations with ring electrodes, we found that regions farther from the electrode were rarely activated) (Fig. 1).

We implemented extracellular stimulation of axons in three steps. First, the steady-state electric field was calculated using the COMSOL finite element model. The steady-state potential values at the center of each axon compartment was then extracted using interpolation based on the positions of the electrodes and axons. Second, stimulation signals were generated according to the pulse width and amplitude, and these signals were multiplied by the potential at the center of each compartment to create the stimulation array for each compartment. Finally, the NEURON "play" method was employed to apply these arrays to the MRG axon model, with simulations conducted at a time step of 5 μs. The minimum stimulation amplitude required to activate the axon, known as the activation threshold, was determined using a binary search algorithm [23]. An axon was considered activated if any node of Ranvier generated action potentials at the same frequency as the stimulation signal [20]. Specifically, an action potential was defined as a transmembrane potential increase above -20 mV at any node of Ranvier [28]. The simulation duration

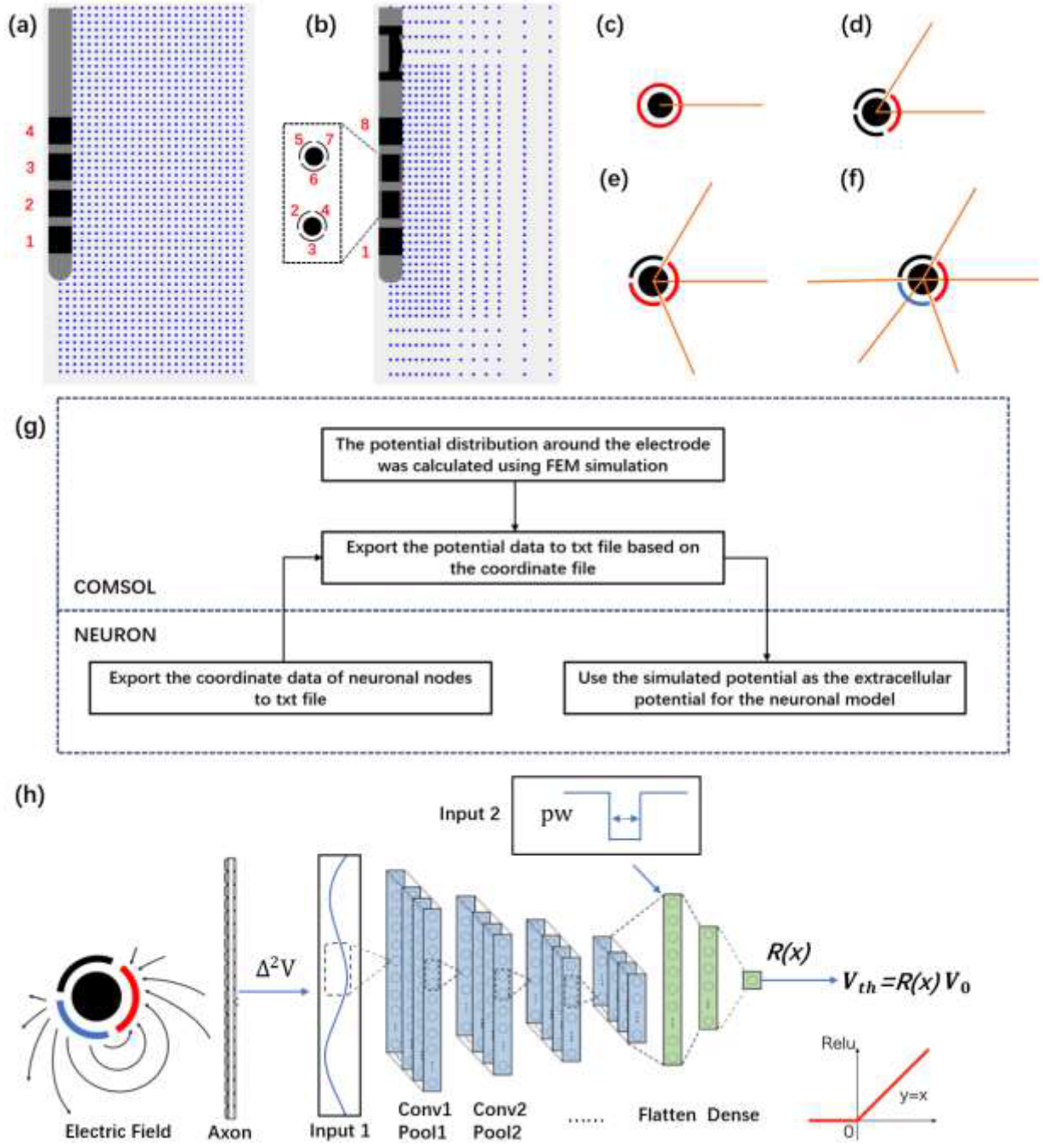


Fig. 1. Method diagram. (a) Arrangement of axons with the Lead-1 ring electrode, the red number is the contact number, and each blue point is the axon perpendicular to the paper surface. (b) Arrangement of axons with the Lead-2 directional electrode. (c) For the ring electrode, axons are calculated in only one plane. (d) If the active contacts of the directional electrode are on the same side, two computation planes are selected, e.g., Lead-2_monopolar_2-C+, Lead-2_bipolar_2-5+, where 2- represents the contact number 2 is negative, C+ represents the implantable pulse generator housing is positive. (e) If the active contacts of the directional electrode are on different sides and under monopolar stimulation, three computation planes are selected, e.g., Lead-2_monopolar_2-3-C+. (f) If the active contacts of the directional electrode are on different sides and under bipolar stimulation, five computation planes are selected, e.g., Lead-2_bipolar_2-3+. The color of the electrode contacts represents different states: black indicates inactive contacts, red indicates negative contacts, and blue indicates positive contacts. (g) The combination of COMSOL and NEURON. (h) Model architecture of CNN.

for the MRG axon model was set to 250 ms [20], with a fixed stimulation frequency of 130 Hz. Thus, the criterion for axon activation was that any node of Ranvier should generate 32 action potentials at the same frequency within the 250 ms following a 130 Hz DBS stimulation.

We computed the response data of the MRG axon model for 29 configurations (Table I), including both ring electrodes and directional electrodes, with stimulation pulse widths ranging from 30 to 450 μs and a frequency of 130 Hz. In total, approximately 650,000 data points were generated (with only

TABLE I
RANGE OF STIMULATION SETTINGS AND ELECTRODE CONFIGURATIONS USED IN THIS STUDY

Items	Range
Electrode Type	Lead-1, Lead-2
Frequency	130 Hz
Pulse-widths	30,50,60,70,80,90,100,110,120,150,200,250,350,450 μs
Amplitudes	0~10 V
Contact configurations	Lead-1(6 monopolar, 10 bipolar), Lead-2(6 monopolar ,7 bipolar)

TABLE II
TOP PERFORMING CNN HYPERPARAMETER OPTIMIZATION

Hyperparameter	Searched Values	Optimum Parameter
Number of convolutional layers	1~7	3
Number of convolutional filters for each layer	1~64	25-17-6
Convolutional kernel size for each layer	3~20	7-15-17
Number of dense layers	1~4	1
Number of dense units	16~320	256
Activation function	ReLU, Tanh, Sigmoid	ReLU
Learning rate	0.0001, 0.0003, 0.001, 0.003, 0.01	0.001
Dropout rate	0, 0.2, 0.5	0
L2 regularization	0, 0.0001, 0.001, 0.01	0

120,000 data points having activation thresholds <10 V being used). Each data point included the axon stimulation input (the potential distribution along the nodes of Ranvier in the axon direction and the stimulation pulse width) and the corresponding output (activation threshold).

C. CNN-Based Predictor

1) Model Architecture: The purpose of the CNN model is to predict the response of a single MRG axon model quickly and accurately. The input to the CNN model includes two features of the extracellular stimulation of the axon: (1) The first input is the pulse width of the stimulation; (2) The second input is the second spatial derivative of the extracellular potential at the nodes of Ranvier along the axon. As suggested by McNeal et al. [29], the neuronal response to electrical stimulation is not directly related to the extracellular voltage or electric field but rather to the second spatial derivative of the extracellular voltage distribution. The second derivative of the potential along each axon is represented as a one-dimensional array. In the CNN model, a series of one-dimensional convolutional layers are applied to process this data. Each convolutional layer is followed by a max-pooling layer. After several layers of convolution, the data is flattened and combined with the stimulation pulse width for further processing in fully connected layers. After passing through several hidden layers, the final output is the activation threshold.

2) Training Method: The entire machine learning process, including the creation, training, optimization, and testing of the CNN, was carried out using TensorFlow (version 2.15.0) in the Python (version 3.9.18) environment. All data was randomly shuffled and divided into training, validation, and test sets in a ratio of 6:2:2. To prevent overfitting, an early stopping strategy was employed: training was halted when the mean absolute error (MAE) on the validation set did not improve for ten consecutive epochs, and the best model weights were restored. The training pulse width range was 30–450 μs, with pulse width linearly mapped to a range of 0–1 before being input into the CNN. The model was trained only within the activation threshold range of 0–10 V, as stimulation amplitudes above 10 V are typically not used clinically. Training was performed using TensorFlow's default compilation and fitting methods, with the Adam optimizer and mean absolute percentage error as the loss function.

The specific structure of the CNN model was determined through hyperparameter tuning. A hyperparameter search was conducted within a range of model parameters of interest (e.g., the number of convolutional layers, the number and length of convolutional kernels, the number of hidden layers, learning rate, activation function, etc.). The hyperparameter search was carried out using the Hyperband tuner from the Keras Tuner library, following a stepwise elimination strategy. In the initial phase, a large number of models were trained for several epochs, and the MAE on the validation set was used as the comparison metric. The best-performing half of the models were selected for the next round of training, and this process was repeated until the best model was obtained. The Table II shows the range and final results of the hyperparameter search.

D. Performance Analysis

For comparison, we trained a total of four models, including two CNNs and two ANNs. The ANN models were identical to the CNNs in all aspects of training, optimization, and testing, except for the absence of convolutional layers. Each model was tested with two types of input: both included the second derivative of the potential and the stimulation pulse width, differing only in whether they included the electric potential data itself. After training, the models were tested

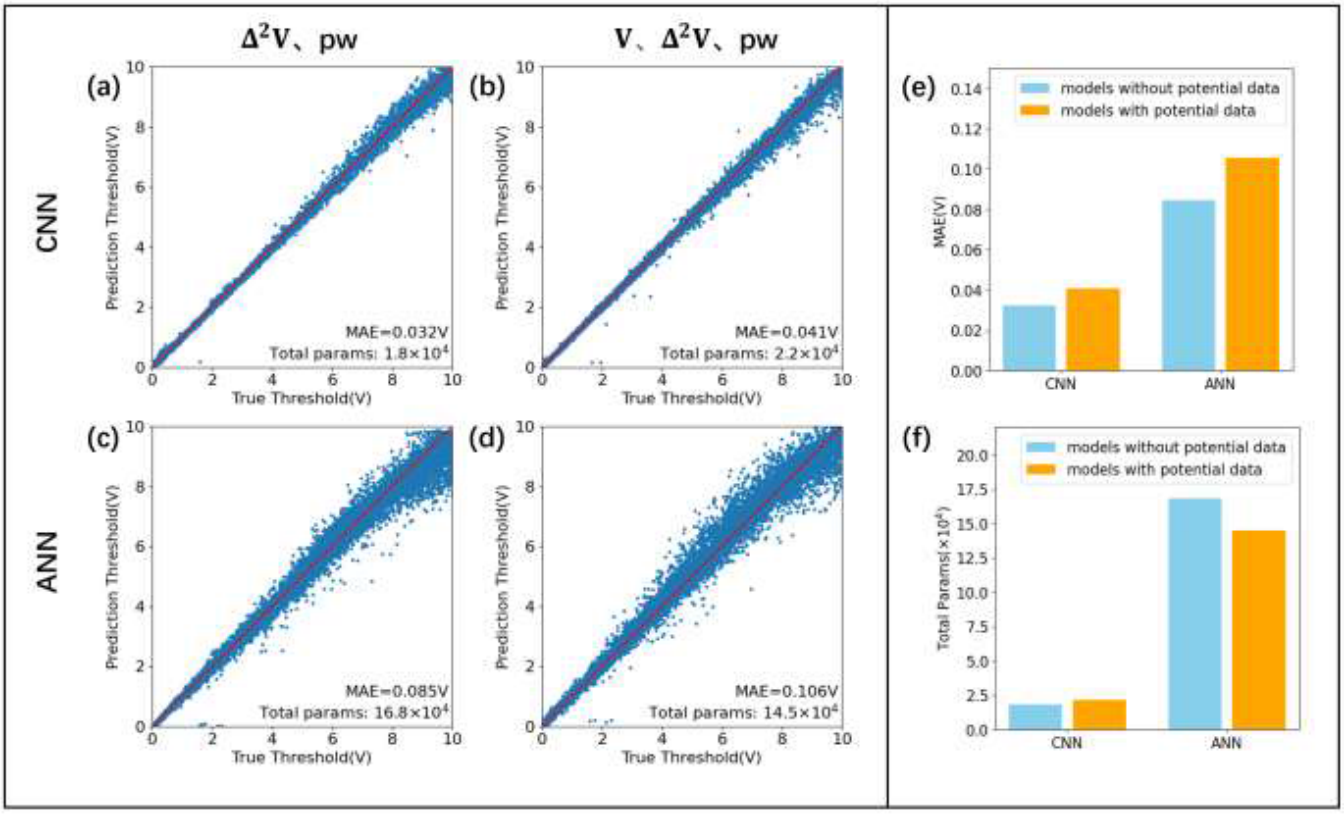


Fig. 2. Test results of 4 models. (a) CNN without electric potential data. (b) CNN with electric potential data. (c) ANN without electric potential data. (d) ANN with electric potential data. (e) Comparison of the mean absolute error. (f) Comparison of the total params.

using the data from the test set. The activation thresholds predicted by the neural networks were directly compared to the standard results from the MRG model using mean absolute error (MAE) and mean absolute percentage error (MAPE) as evaluation metrics. We trained the neural network on an NVIDIA GeForce RTX 4060 Laptop GPU with 24 GB of memory (8 GB dedicated memory + 16 GB shared memory).

Furthermore, we evaluated the performance of the best-performing CNN model in calculating the VTA. MATLAB software was used to plot the distribution maps of axonal activation thresholds at each grid point under different electrode stimulations, and the VTA range was then delineated based on the stimulation amplitude.

The similarity between the VTAs generated by the CNN method and the AFmax method [20] compared to the standard axon model results was quantified using the Dice coefficient. The Dice coefficient is defined as $2|A \cap B|/(|A| + |B|)$, where A is the VTA derived from the axon model, and B is the VTA from the method being compared (either CNN or AFmax). The Dice calculation was performed by identifying the shared and unique activation grid points between the two VTAs across all computed planes. We evaluated various configurations and stimulation parameters for both ring and directional electrodes, with stimulation amplitudes ranging from 0 to 10 V and pulse widths ranging from 30 to 450 μs.

Finally, we compared the runtime of each model for calculating activation thresholds for different numbers of axons. The runtime statistics were performed on a system equipped with an Intel Core i9-13900H processor (2.6 GHz, 14 cores) and 32 GB of RAM.

III. Results

A. Predicting Activation Thresholds

Among the four neural network models, the CNN model without the electric potential data predicted activation thresholds most accurately (Fig. 2). The MAE values for this model on the training, validation, and test sets were 0.0391 V, 0.0374 V, and 0.0322 V, respectively, while the MAPE values were 0.827%, 0.747%, and 0.812%. Compared to the ANN models, the CNN models have fewer parameters, resulting in lower training and prediction time costs, as well as better overall performance. Compared to models without the electric potential data, the models with electric potential data had double the input parameter dimensions and were more complex, but this did not result in better outcomes.

B. Performance in Predicting VTA

The distribution of axonal activation thresholds and the VTA predicted by the best-performing CNN model closely resemble the standard results obtained using the axon model method (Fig. 3).

We compared the performance differences between the CNN model and the widely used AFmax method, where AFmax refers to the maximum value of the second derivative

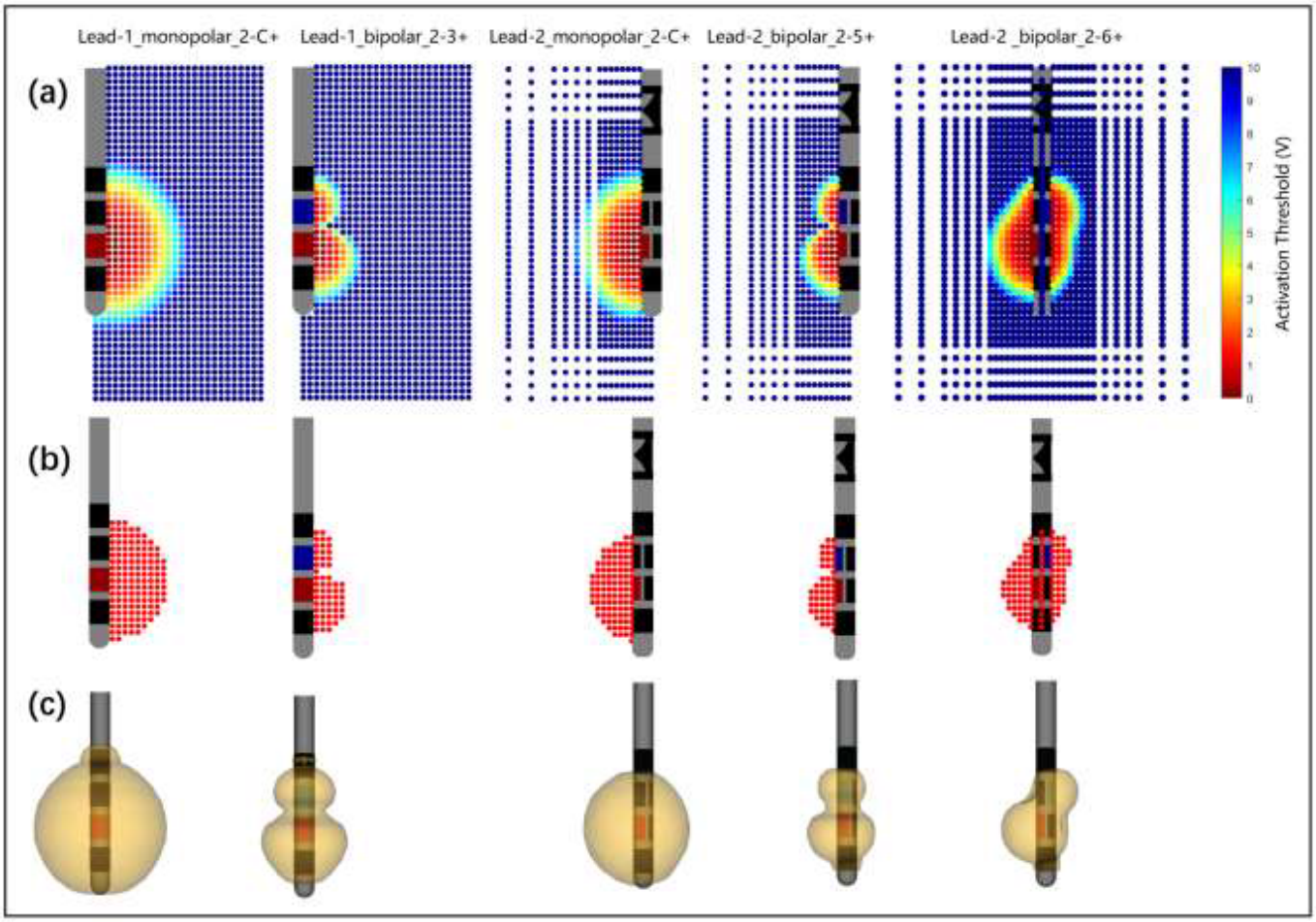


Fig. 3. VTA diagram. (a) Map of axon threshold predicted by CNN model. (b) Activated axons at 3.0V stimulation amplitude, 90μs pulse width, and 130Hz stimulation frequency. (c) VTA surface calculated by the axon model method in Duffley [14].

of the potential at all central compartments. Across the examined electrode configurations, stimulation pulse widths, and stimulation amplitudes, the CNN consistently outperformed the AFmax method (Fig. 4). The AFmax method performed poorly under certain parameter conditions, such as when the stimulation pulse width approached 450 μs or the stimulation amplitude was less than 2 V. In contrast, the CNN exhibited stable performance, closely approximating the results of the standard method.

The CNN model demonstrated exceptionally high performance across the 29 electrode configurations it was trained on. Given that the training data came from these configurations, such performance is not surprising. To better assess the model's generalization capability, we further tested its predictive accuracy on configurations not included in the training dataset, such as the current mode of Lead-1, electrodes with different contact spacings like Lead-3 and Lead-4 (where Lead-3 is a ring electrode and Lead-4 is a directional electrode, both with a 1.5 mm contact spacing, otherwise identical to Lead-1 and Lead-2), and additional directional contact combinations of Lead-2. The model continued to perform well across these scenarios.

C. Comparison of Runtime

The CNN model incorporates the second-order derivatives of the extracellular potentials at all Ranvier nodes of the axon, providing more detailed and accurate information compared to the AFmax method. When compared to the standard axon model method, the CNN model achieves a computational speedup of five orders of magnitude (Fig. 5).

IV. Discussion

A. Patterns of Axon Activation

Warman et al. [30] pointed out that the response of the Ranvier node, where the axon action potential initiates, is primarily influenced by two factors: the activation function of that node, which is the second spatial derivative of the potential, and the current generated by surrounding nodes. The contributions of nodes beyond five nodes away from the point of interest are negligible. Peterson [22] achieved high-precision activation threshold calculations using a weighted sum of the second derivatives of the extracellular potential, where the node closest to the stimulation source has the highest weight, and the weight decreases as the distance increases. This window, formed by the Ranvier node at the maximum of the second derivative of the potential and several surrounding nodes, can be used to precisely calculate the activation threshold, with the central node having the highest weight and the weight gradually decreasing outward. However, to calculate this accurately, it is necessary to locate this window. Under the electric field formed by the ring electrode, this window is located at the center of the axon (for monopolar cases,

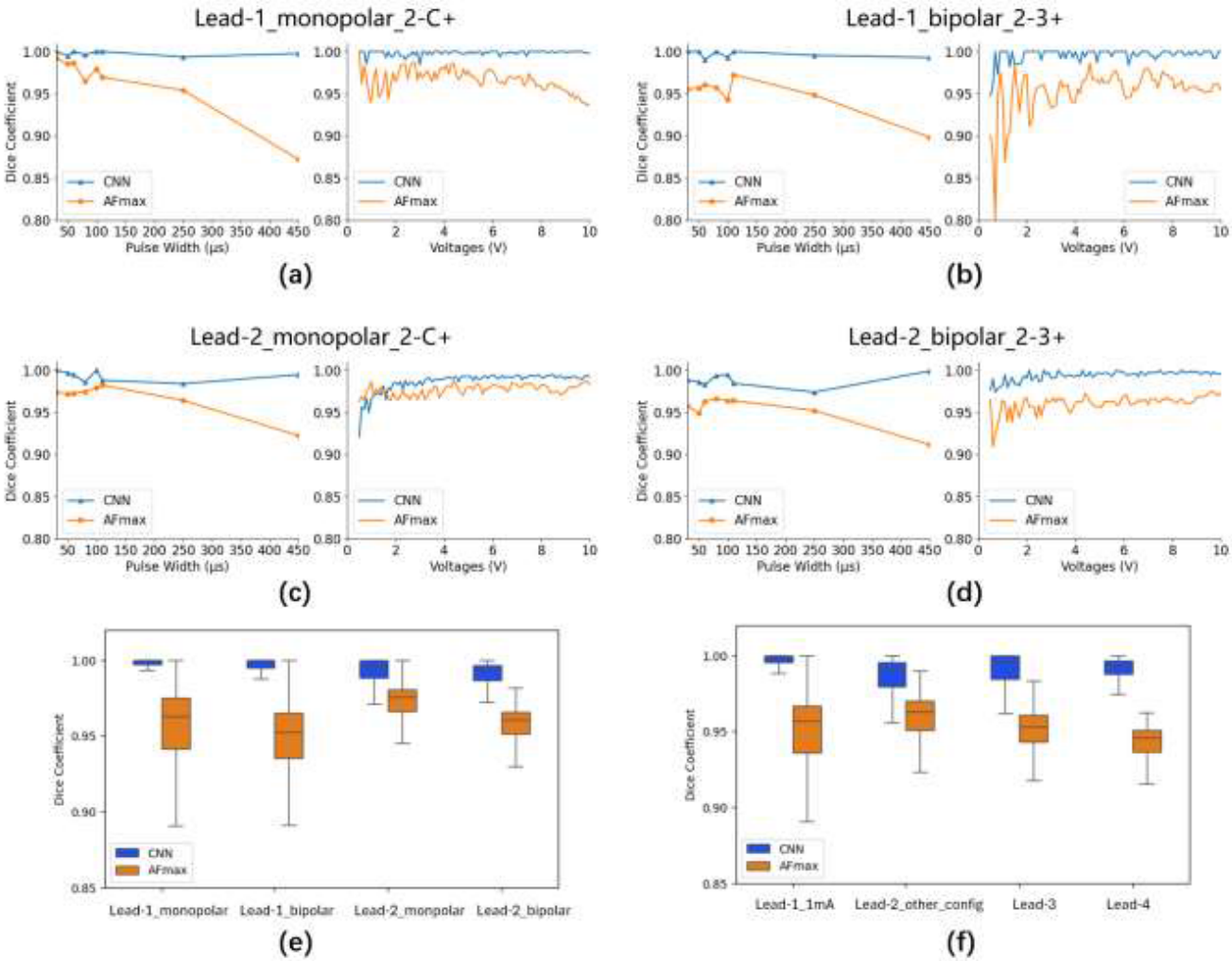


Fig. 4. Performance comparison between CNN model and AFmax method under various electrode configurations and stimulation parameters. (a) Lead-1_monopolar_2-C+. (b) Lead-1_bipolar_2-3+. (c) Lead-2_monopolar_2-C+. (d) Lead-2_bipolar_2-3+. (e) Statistical analysis of the performance of the two methods under the above 29 electrode configurations. (f) Statistical analysis of the performance of the two methods in different electrode configurations other than the above data set.

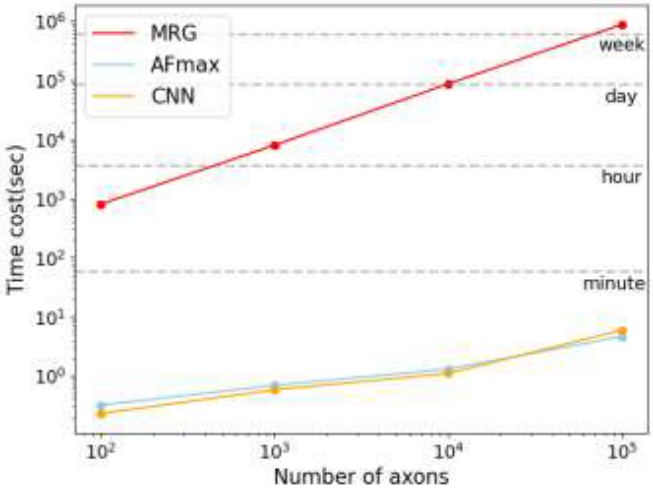


Fig. 5. Time cost of calculating activation thresholds for different models.

the location differs slightly for bipolar cases). In the slightly more complex electric field environment created by directional electrodes, the position of this window is not fixed. CNNs, which have translational invariance, are well-suited to address this issue as they can learn the characteristics of the window that triggers axon activation from large amounts of data, regardless of where this window is located along the axon.

B. Rationale for Using the MRG Model to Predict the Effects of DBS

The basal ganglia, including regions such as the subthalamic nucleus and globus pallidus, are the primary targets of DBS [31]. Neurons in these areas are of various types, including myelinated and unmyelinated neurons [32]. In this study, we focus primarily on the axons of myelinated neurons. Clinical, experimental, and theoretical studies suggested that the therapeutic response to DBS is mainly associated with the axonal activation [33], [34] that electrical stimulation typically activates myelinated axons first, while unmyelinated axons are less likely to be activated [35], [36]. For example, myelinated nerve fibers are most probably the elements in STN, VIM and GPi targeted by DBS, and the effect of stimulation may be mediated by both afferents and efferents of these nuclei [37], [38].

TABLE III
COMPARISON WITH OTHER SIMILAR STUDIES

Study	Method	Calculating parameter	Electrode configuration	Brain model	Computational accuracy	Computational cost	Characteristic
This study	CNN predicts the axon activation threshold	Δ^2V, pw	Ring / directional	Homogeneity, isotropy	Dice 0.99 MAE 0.034	0.6s / 1250 axons MRG 0.008%	Complete electrode configuration
Golabek 2023 [23]	ANN predicts axon response(regression/ classification)	V, Δ V, Δ^2V, pw	Monopolar of ring electrode	Homogeneity, isotropy	MAE 0.037	0.07s / 288 axons MRG 0.004%	Non-straight axon
Duffley 2019 [20]	AFmax	The maximum of Δ^2V, pw	Ring / directional	Homogeneity, isotropy	Dice 0.9-0.95	0.7s / 1250 axons MRG 0.009%	Unconstant performance
Panche 2017 [19]	Gaussian process classifier	V, Ranvier node-electrode distance	Ring	Isotropy / anisotropy	Error 0.1	267.7s / 500 axons MRG 10%	Poor accuracy and cost
Astrom 2015 [18]	Electric field norm	L^2 norm of electric field	Monopolar of ring electrode	Homogeneity, isotropy	Ordinary	Minimum	Suitable only for monopolar of ring electrode
Chaturvedi 2014 [17]	ANN predicts the size of an elliptical VTA	Pw, amplitude, impedance, fiber diameter	Ring	Homogeneity, isotropy	Error 0.13±0.03	Low	Double the bipolar configuration error
Peterson 2011 [22]	Make the electric field feature predict activation threshold	The weighted sum of V, pw, fiber diameter	Point source electrode	Homogeneity, anisotropy	Threshold error 0.7%~45.8%	MRG 0.005%	Multiple fiber diameters

In the table, "V" represents the potential applied to the axon, "ΔV" represents the derivative of the potential along the direction of the axon, "Δ²V" represents the second derivative of the potential along the direction of the axon, and "pw" represents the stimulation pulse width. The computational cost may vary across different computers and is provided for reference only.

The MRG model is currently a gold standard for myelinated axon modeling, which can reproduce a wide range of experimentally measured excitation characteristics due to its introduction of ion channels such as fast Na+ channels, persistent Na+ channels, and slow K+ channels [26]. It is an excellent tool and has been widely used to study the neural responses to DBS [9], [13], [27]. In addition to calculate the activated volume in the target regions, the MRG model can also be used to predict the activation of myelinated fibers in the surrounding nuclei during DBS. It could provide guidance to clinicians to avoid stimulating these fibers to prevent possible side effects [39].

It is worth noting that the ion channels used in the MRG model are simplified representations of those experimentally recorded in actual nerve fibers. These ion channels are strategically selected and modified to excellently reproduce multiple experimentally documented excitation characteristics. They may actually represent the condensation of several different types of channels into single equivalent channels [26].

C. Principles for Selecting Computation Planes

In Fig. 1 (d), (e), and (f), we selected different computation planes for various electrode configurations. Performing calculations for every point in the entire three-dimensional space around the electrode is computationally expensive and involves a significant amount of redundant computation. The selection of computation planes is primarily based on avoiding redundant calculations. For each electrode configuration, the selected computation planes have distinct electric field distributions, ensuring that unique characteristics of different configurations are preserved while reducing the overall computational cost.

For the centrally symmetric electric field of a ring electrode (Lead-1), it is sufficient to calculate axons in only one plane. For the configuration shown in Fig. 1(d), where only one side's contact is active, two computation planes are selected: one along the center and the other along the edge of the active contact. Axons activated near the inactive contacts contribute minimal data. For Fig. 1(e), where two active contacts are on opposite sides and of the same polarity (e.g., both cathodes), the potential distribution is symmetric along the midline between the two contacts, so only three computation planes are needed. For Fig. 1(f), where the two active contacts are on opposite sides but have different polarities (cathode and anode), the potential distribution is asymmetric, requiring five computation planes.

In summary, for each electrode configuration, the selection of computation planes aims to minimize the number of planes, avoid redundant calculations between planes, and retain the unique characteristics of each configuration's electric field distribution.

D. Advantages Compared to Other Models

In Table III, we compare this study with other similar studies. The activation function (AF) method simplifies the information about the extracellular potential distribution of the axon, considering only the maximum value of the second derivative of the potential as the relevant electric field parameter. This simplification is sometimes insufficient to accurately describe the complex nonlinear behavior of the cable model, leading to lower accuracy and instability at high stimulation pulse widths and low stimulation amplitudes. In contrast, the CNN model uses the second derivative of the potential at all Ranvier nodes along the axon as input, thereby considering more effective information. With proper

training, the convolutional neural network can predict the activation threshold of axons with extremely high accuracy and perform consistently across various electrode configurations and stimulation parameters.

The ANN model proposed by Golabek [23] only addresses the precise prediction of axon activation under monopolar stimulation with ring electrodes. It considers parameters such as the potential and its first and second derivatives. However, we believe that using only the second derivative of the potential is sufficient for efficient prediction; introducing additional parameters could bring in unnecessary data and negatively impact the model's performance. Golabek's model selects the initiation point of the axon action potential based on the minimum potential value, which is only correct in the monopolar case. When considering both monopolar and bipolar situations, the initiation point should be at the location of the maximum second derivative of the potential. Compared with ANN, our CNN model incorporates axon response data from a broader range of electrode configurations, enabling precise calculations even under more complex electric fields, such as those from bipolar stimulation or directional electrodes. Currently, the CNN model can be used for real-time VTA calculation, serving as a visual guide for surgical planning and DBS programming, and it has the potential to further predict pathway activation. [20]

E. Study Limitations and Future Directions

The primary drawback of the proposed CNN model is the significant amount of data and time required for training. We incorporated tens of thousands of axon simulation response data under nearly 30 electrode configurations, including ring electrodes and directional electrodes, which incurred substantial time costs. Additionally, finding the optimal model requires a comprehensive hyperparameter search during training to identify the best combination of hyperparameters, a process that takes approximately several hours. On a single CPU, training the optimal model takes about 40 minutes. These upfront time costs are considerable.

Moreover, the training data for the current CNN model is based on electric fields generated by ring electrodes and directional electrodes, with axons arranged orthogonally to the electrodes. This potential distribution data is quite limited, and the model's effectiveness cannot be guaranteed if applied to predict axon responses under more complex or drastically different electric fields. In clinical settings, individual anatomical and physiological differences among patients—such as brain anatomy, electrode placement accuracy, tissue conductivity, and variations in neural fiber orientation—can influence the electric field distribution, thereby affecting the CNN model's predictions of axonal activation.

To further improve the generalization capability of the CNN model and achieve personalized predictions of axonal activation, the following measures can be considered: Building a more comprehensive dataset: Future work could involve generating additional electric field data to simulate axonal responses under various complex electric fields, including as many scenarios as possible (e.g., straight and curved axons). Techniques such as data augmentation, altering electrode positions, simulating different tissue conductivities, and introducing diverse axonal pathways could be employed to create a more robust training set. This comprehensive dataset could support the training of a more "generalized" model. The CNN model should go beyond the assumption of orthogonal axonal arrangements in existing VTA models. By integrating with diffusion tensor imaging (DTI) technology, the model could predict axonal activation while accounting for patient-specific neural fiber orientations. Exploring advanced neural network architectures: Adopting more advanced architectures, such as attention-based models, could further enhance the model's ability to handle complex and personalized electric fields.

V. Conclusion

In this paper, we propose for the first time the use of CNNs to predict the activation thresholds of myelinated nerve fibers under DBS stimulation. The results show that the CNN-based neural fiber activation predictor can rapidly and accurately predict the response of the multi-compartment axon model under different electric field distributions and stimulation pulse widths. Compared to the AF method, it offers higher accuracy, performs stably across various stimulation parameters, and has the potential for real-time prediction and clinical application.

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